

100 AI / ML / DL / LLM PROJECTS

A Project Selection Paper with Top 20 High-Impact Projects Marked

Purpose: This paper lists 100 project ideas across Machine Learning, Deep Learning, Transformers/LLMs, RAG, AI Agents, Multimodal AI, MLOps/LLMOps, AI for Science, Robotics/RL/Embodied AI, and Full-Stack AI. The top 20 are marked with ★.

Important: The ★ mark identifies projects selected as especially deep, research-oriented, or production-oriented within this list. It is not a guarantee of hiring outcome.

Machine Learning

1	★ Customer Churn Prediction Platform
2	Fraud Detection & Real-Time Risk Scoring
3	Credit Risk & Explainable Lending System
4	Recommendation Engine
5	Demand Forecasting Platform
6	Predictive Maintenance System
7	Dynamic Pricing Engine
8	Search Ranking & Learning-to-Rank System
9	Customer Lifetime Value Prediction
10	ML Model Monitoring, Drift & Retraining Platform

Deep Learning

11	Computer Vision Image Classification Platform
12	Real-Time Object Detection System
13	Semantic Segmentation Platform
14	Medical Image Analysis Research System
15	Deep Time-Series Forecasting Platform
16	Deep Anomaly Detection System
17	Neural Recommender System
18	Speech Emotion & Audio Classifier
19	Image Captioning System
20	Deep Learning Experiment Platform

Transformers & LLMs

21	★ Transformer Language Model From Scratch
22	★ Small Language Model Pretraining Lab
23	★ Domain-Specific Instruction-Tuned LLM
24	LoRA/QLoRA Fine-Tuning Laboratory

25	★ LLM Inference Optimization & Quantization Platform
26	★ LLM Evaluation & Benchmarking Platform
27	Long-Context LLM System
28	★ Multimodal LLM Application
29	LLM Safety & Guardrails Laboratory
30	Multi-Model LLM Gateway

RAG & Knowledge

31	★ Enterprise Multimodal RAG Platform
32	★ Real-Time Corrective Agentic RAG (CRAG)
33	★ GraphRAG Knowledge Engine
34	Research Paper Intelligence System
35	Text-to-SQL RAG Agent
36	Document Intelligence Pipeline
37	RAG Evaluation & Regression Harness
38	Semantic Cache for LLM/RAG
39	Enterprise Knowledge Search Engine
40	Multimodal Document Question Answering

AI Agents & Automation

41	Autonomous AI Research Scientist
42	★ Autonomous Software Engineer
43	★ Multi-Agent AI Research Platform
44	★ MCP-Powered Coding Agent
45	★ Browser / Computer-Use AI Agent
46	★ Persistent-Memory Personal AI Agent
47	Self-Improving Reflexion Agent
48	Hierarchical Multi-Agent Supervisor
49	AI Task Planning & Execution System
50	Voice AI Support Agent

Multimodal AI

51	Real-Time Voice Interviewer
52	Vision-Language Document Analyst
53	Video Understanding & Summarization AI
54	Multimodal Meeting Intelligence System
55	Visual Web Research Agent
56	CLIP-Based Image Search

57	Chart & Table Reasoning Assistant
58	Audio-Visual Event Detection System
59	Multimodal Education Tutor
60	Multimodal Medical Research System

MLOps & LLMOps

61	AI Model Registry & Experiment Tracking Platform
62	★ LLM Observability & Tracing Platform
63	AI Cost Optimization Autopilot
64	LLM Prompt Versioning & A/B Testing Platform
65	Model Regression Detection CI/CD System
66	AI Failure Forensics Platform
67	AI Data/Model Drift Monitoring System
68	LLM Red-Team Evaluation Suite
69	★ Production AI Gateway with Rate Limits
70	AI Quality Dashboard with Golden Datasets

AI for Science

71	★ AI-for-Science Discovery Platform
72	Protein Structure Prediction Research System
73	Drug Discovery Candidate Ranking
74	Materials Property Prediction System
75	Scientific Literature Knowledge Graph
76	Climate Forecasting AI
77	Physics-Informed Neural Network Platform
78	Bioinformatics Sequence Intelligence System
79	Automated Research Literature Synthesizer
80	Scientific Hypothesis Generation & Verification Agent

Robotics, RL & Embodied AI

81	Reinforcement Learning Game Agent
82	Robot Navigation & Path Planning Simulator
83	Vision-Based Robotic Manipulation Simulator
84	★ World Model + Embodied AI Simulator
85	Autonomous Drone Perception Simulator
86	Multi-Agent Reinforcement Learning Lab
87	Robot Task Planning Agent
88	Simulated Warehouse Robot Fleet

89	Humanoid Motion Planning Research Platform
90	Embodied AI Household Task Agent

Full-Stack AI

91	AI Personal Operating System
92	AI Knowledge Workspace
93	★ AI Coding & Research IDE
94	AI Finance & Expense Intelligence SaaS
95	AI Healthcare Document Intelligence SaaS
96	AI Education Copilot Platform
97	AI Customer Support Automation SaaS
98	AI Developer Productivity Platform
99	AI Security Operations Assistant
100	Enterprise AI Operating System

TOP 20 PROJECTS — ★

1	Customer Churn Prediction Platform
21	Transformer Language Model From Scratch
22	Small Language Model Pretraining Lab
23	Domain-Specific Instruction-Tuned LLM
25	LLM Inference Optimization & Quantization Platform
26	LLM Evaluation & Benchmarking Platform
28	Multimodal LLM Application
31	Enterprise Multimodal RAG Platform
32	Real-Time Corrective Agentic RAG (CRAG)
33	GraphRAG Knowledge Engine
42	Autonomous Software Engineer
43	Multi-Agent AI Research Platform
44	MCP-Powered Coding Agent
45	Browser / Computer-Use AI Agent
46	Persistent-Memory Personal AI Agent
62	LLM Observability & Tracing Platform
69	Production AI Gateway with Rate Limits
71	AI-for-Science Discovery Platform
84	World Model + Embodied AI Simulator
93	AI Coding & Research IDE

Suggested progression for building these projects

1. Foundation: Machine Learning → Deep Learning → PyTorch → Transformers.
2. LLM core: Transformer from scratch → Small Language Model → Fine-tuning → Inference and quantization.
3. RAG: Enterprise Multimodal RAG → GraphRAG → CRAG → RAG evaluation.
4. Agents: tool calling → LangGraph → MCP → multi-agent research → coding/browser agents.
5. Production: FastAPI → PostgreSQL/vector database → Docker → CI/CD → observability → cloud.
6. Research: AI for Science / World Models / Reinforcement Learning / Embodied AI.

Project-depth note: the supplied AI Engineering Projects Guide recommends choosing 2–3 projects and building them exceptionally well rather than building many shallow projects. It emphasizes evaluation, cost optimization, observability, CI/CD, retrieval, fine-tuning, and multi-agent orchestration as signals of serious AI engineering work.